

Ball action spotting

C6 - Video Analysis



Team 5

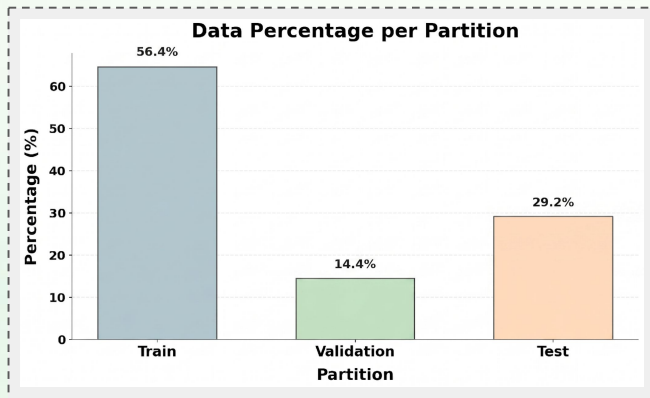
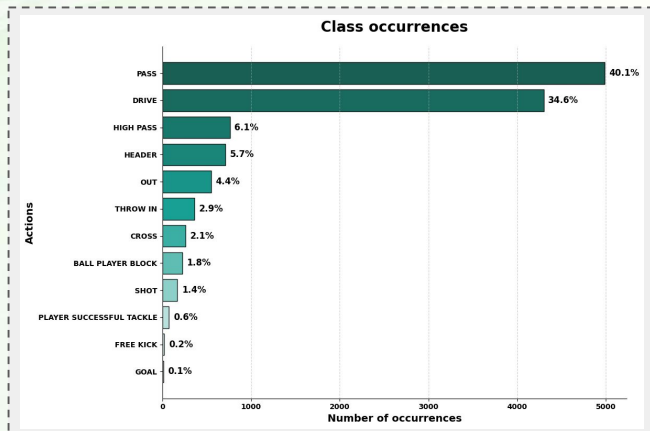
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Table of Contents

1	Dataset
2	Challenges
3	Past Weeks
4	Main Objectives
5	Approach & Baseline
6	Experiments

7	Best Model
8	Quantitative
9	Qualitative
10	Conclusions
11	Future Work



*Measured in frames

Data: 7 soccer games

Split: 4 Training / 1 Validation / 2 Testing

Classes:

- **PASS**
- **DRIVE**
- **HEADER**
- **HIGH PASS**
- **OUT**
- **CROSS**
- **THROW-IN**
- **SHOT**
- **BALL PLAYER BLOCK**
- **PLAYER SUCCESSFUL TACKLE**
- **FREE KICK**
- **GOAL**

Original Paper

SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos

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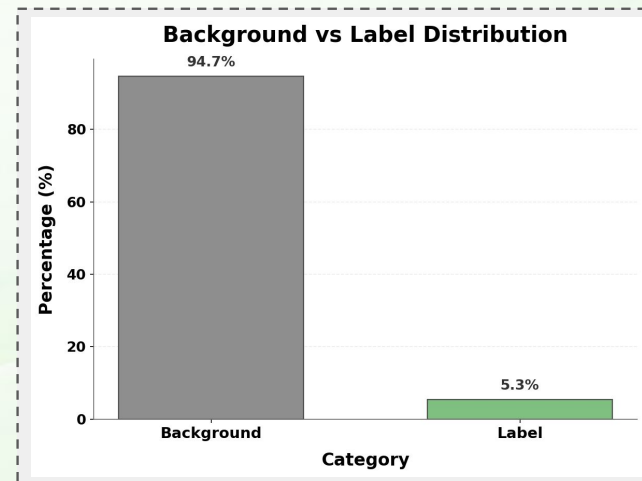
Abstract

In this paper, we introduce SoccerNet, a benchmark for action spotting in soccer videos. The dataset is composed of 500 complete soccer games from six main European leagues, covering three seasons from 2014 to 2017 and a



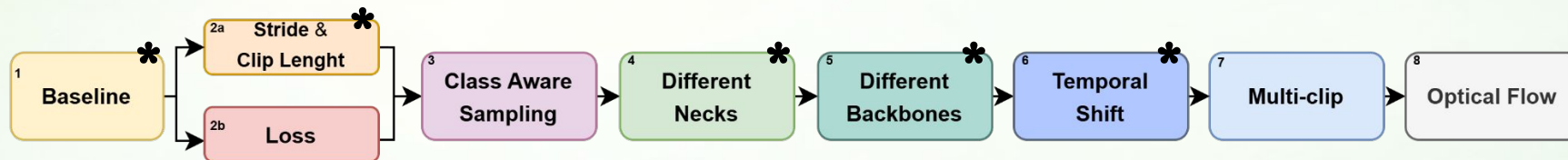
Challenges:

- Difficulties in **knowing exactly which class** we are trying to **predict**.
- Most of the frames in the dataset are **Background**.
- **Dataset imbalance**



Past Weeks

Week 5 - Ball Action Classification



Metrics			
<i>mAP10</i>	<i>mAP12</i>	<i>Params (M)</i>	<i>MACs (G)</i>
0,31	0,28	2.778	18.105

What did bring us
from the baseline to
our best model?

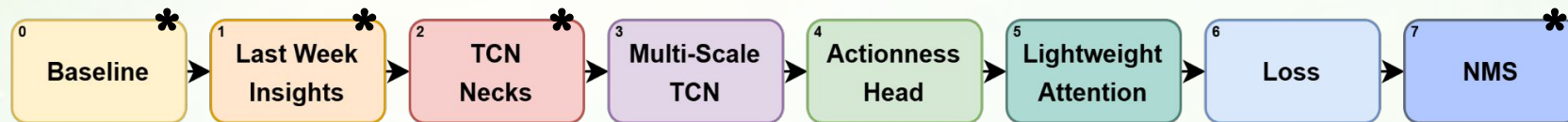
Best Model

Metrics			
<i>mAP10</i>	<i>mAP12</i>	<i>Params (M)</i>	<i>MACs (G)</i>
0,55	0,51	10.789	71.720

- + RNY008
- + TCN-MaxPool (3-layer, kernel 3, dropout 0.2)
- + Temporal Shift
- + Stride 4

Past Weeks

Week 6 - Ball Action Spotting



Metrics			
<i>mAP10</i>	<i>mAP12</i>	<i>Params (M)</i>	<i>MACs (G)</i>
0,10	0,08	2.778	18.105

What did bring us
from the baseline to
our best model?

Best Model

Metrics			
<i>mAP10</i>	<i>mAP12</i>	<i>Params (M)</i>	<i>MACs (G)</i>
0,35	0,34	10.789	71.720

+ RNY008

+ TCN-MaxPool

+ NMS

(3-layer, kernel 3,
dropout 0.2)

(Hard, window 10, threshold 0.03,
mean smoothing (window 3))



Key Insights!!



– More **temporal resolution** → More **context** for the predictions.



– Techniques to solve the **Class Imbalance** do not work as expected.



– **Higher complexity** → **Higher performance**.



– **TCN** work fine... But **X3D with GRU** works **better**.



Main Objectives

-  Implement **temporal aggregation** using **UNet** or DETR-like architectures.
-  Evaluate **model performance** using different **mAP tolerances** (0.5s and 1.0s).
-  **Refine** the **best-performing model** based on **findings** from previous weeks.

Approach & Baseline

Approach

General:

- **Iterative** experimental procedure
- **Best checkpoint selection** → mAP10 tolerance 1s
- **Epochs** → 3 warmup + 20
- **LR** → 8e-4
- **Early-stopping** → patience = 6
- **Stride**→2 **Clip Len**→50

Experimental pipeline:

1. Create a new setup with **X3D-M**
2. Add a 2-layer bidirectional **GRU**
3. Implement a **U-Net-like** architecture without GRU
4. Add a 2-layer bidirectional **GRU** to the U-Net
5. Evaluate different **clip lengths** and **strides**
6. Replace MaxPool bottleneck with **AvgPool**
7. Perform **GRU hyperparameter search**
 - # layers
 - hidden dimensions)
8. Add **TGLS**

Best Model last week

TCN

RNY008

NMS

(3-layer, 1-2-4 dilation,
kernel 3, dropout 0.2)

Tolerance 1s		Tolerance 0.5s	
mAP 10	mAP 12	mAP 10	mAP 12
35.5	34.2	30.2	29.7

Experiments

Experiments 1 & 2 – X3D Baseline

We start by doing some initial experiments to check if everything is correct.

Experimental Setup

1. X3D (embedding dim 192)
2. X3D (embedding dim 192) + Bi-GRU (2-layer)

Expectations

- **Sanity check** using only X3D.
- Reproduce the results of **Team 2** from the past week by adding **GRU**.

Results & Insights

- We achieved practically the **same results** as Team 2, therefore our baseline is improved.
- **Adding temporal information** is key to have better results.
- **Bi-GRU** increases the performance.

	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
Best Model W6	35.5	34.2	30.2	29.7
1 - X3D Base	41.3	36.7	36.3	32.5
2 - X3D + Bi-GRU	45.3	44.8	40.0	40.4

Experiments

Experiment 3 & 4 - X3D + UNet (+ Bi-GRU)

First experiments using UNet-like architecture and also including Bi-GRU.

Experimental Setup

UNet-like architecture

- 3 { Encoder: X3D (4 stages, embedding dim 192)
 Bottleneck: Max Pooling (kernel size 2, stride 2)
 Decoder: HandMade (4 stages + skip connections)
- 4 - Adding Bi-GRU (2-layer)

Expectations

- Only using UNet **won't yield better results**.
- Using **Bi-GRU** with **UNet** will **increase the performance** of Experiment 2.

Results & Insights

- Using **UNet-like** architecture and **Bi-GRU** does **not improve** the results for mAP10.
- However, **Bi-GRU** results **essential** to have better results.
- **Max Pooling** too aggressive.
- Despite didn't work as expected, Exp. 4 will be the base for the next experiments.

	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
2 - X3D + Bi-GRU	45.3	44.8	40.0	40.4
3 - X3D + UNet	39.1	33.7	34.8	30.1
4 - X3D + UNet + Bi-GRU	43.7	41.7	40.1	38.7

Experiments

Experiments 5 – Stride & Clip Length

Max Pooling is too aggressive. Perhaps with more context per clip could be better.

Experimental Setup

UNet-like architecture +

- Stride → 2 & Clip Length → 50 (Exp. 4)
- Stride → 4 & Clip Length → 50
- Stride → 2 & Clip Length → 100
- Stride → 4 & Clip Length → 100

Expectations

- Adding more **context** to the model can help to increase the results.
- **Clip Length** will be more important than **Stride**.

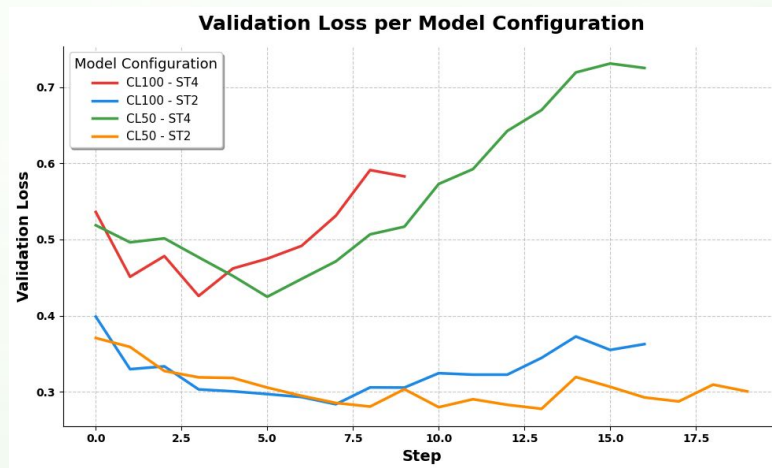
	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
4 - Stride 2 CL 50	43.7	41.7	40.1	38.7
5 - Stride 4 CL 50	43.3	42.2	31.7	32.5
6 - Stride 2 CL 100	43.8	39.4	39.4	35.7
7 - Stride 4 CL 100	45.3	43.0	35.4	34.8

Results & Insights

- **Stride 4** and **Clip Length** of **100** improves the results notably.
- The base experiment (Exp. 4) dominates in tolerance of **0.5s**
- We will keep using the **Stride 2** and **Clip Length** of **50**.

Experiments 5 – Stride & Clip Length

The justification of why we keep using the configuration Stride 2 and CL 100 is the following...



- **Overfitting** in the early epochs.
- **Instability** in the mAP10 plots.
- The base experiment (Exp 4 - CL 50 ST2) shows **stability** throughout the epochs.

Experiments 6 – Average Pooling

Since using other clip len or stride doesn't help, we try using a less aggressive pooling such as average pooling.

Experimental Setup

UNet-like architecture

- Encoder: X3D (4 stages, embedding dim 192)
- Bottleneck: Max / Avg Pooling (kernel size 2, stride 2)
- Decoder: HandMade (4 stages + skip connections)
- Bi-GRU (2-layer)

	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
4 - X3D + UNet + Bi-GRU (MAX)	43.7	41.7	40.1	38.7
8 - X3D + UNet + Bi-GRU (AVG)	45.8	42.7	40.8	38.5

Expectations

- **Soft Polling improves the results** by companting frames less aggressively.

Results & Insights

- **Average Pooling** dominates in **all** the metrics (meaningless difference in tolerance 0.5 mAP12).
- As expected, we can consider that Average Pooling **keeps more information** although the embeddings are reduced.

Experiments 7 – GRU Hyperparameter search

Since GRUs have proven effective, we explore their optimal configuration.

Experimental Setup

For both **Max** and **Avg** aggregation:

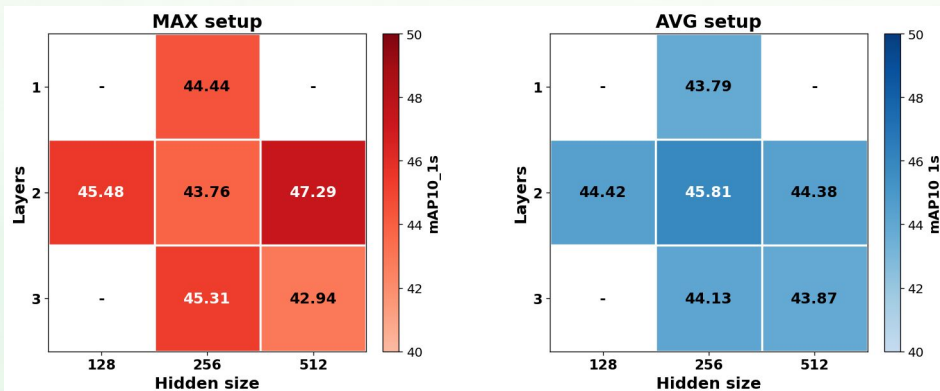
		Hidden Size		
		128	256	512
Layers	1		X	
	2	X	X	X
	3		X	X

Expectations

- Tuning GRU hyperparameters will **boost performance**.
- There will be a **trade-off** between increased model **complexity** and the **risk of overfitting**.

Results & Insights

- **Best setup**: MAX, 2 layers, hidden size 512.
- **AVG limits GRU learning** - most setups perform similarly.
- MAX shows **higher variability**, achieving both **best** and **worst** results.
- The best scores for MAX and AVG are achieved with **different configurations**.



Experiments 8 – Nearly Hit Predictions & Non-Maximum Suppression

After finding the best model so far, we test the if a better supervisory signal for nearly-hit predictions can improve the results.

Experimental Setup

Best model (X3D + Max Pooling + GRU)

+ Temporal Gaussian Label Smoothing (TGLS)

(sigma: 0.55 + window: 5)

+ Non-Maximum Suppression (NMS)

(type: hard, thresh: 0.03,
smoothing: mean, smoothing window: 3)

Expectations

- **TGLS** can **improve** the performance since it penalizes in different way the predictions.
- **NMS** should **increase** the performance since it worked the past week.

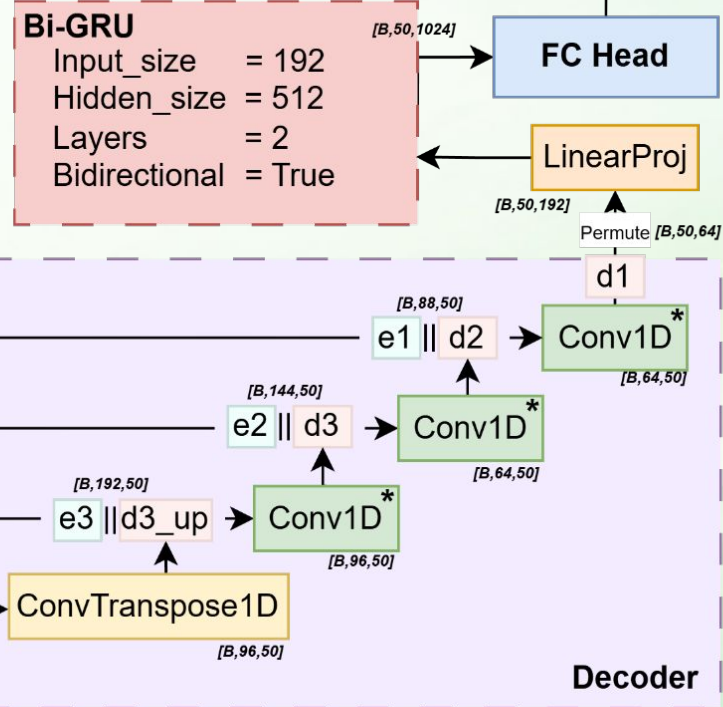
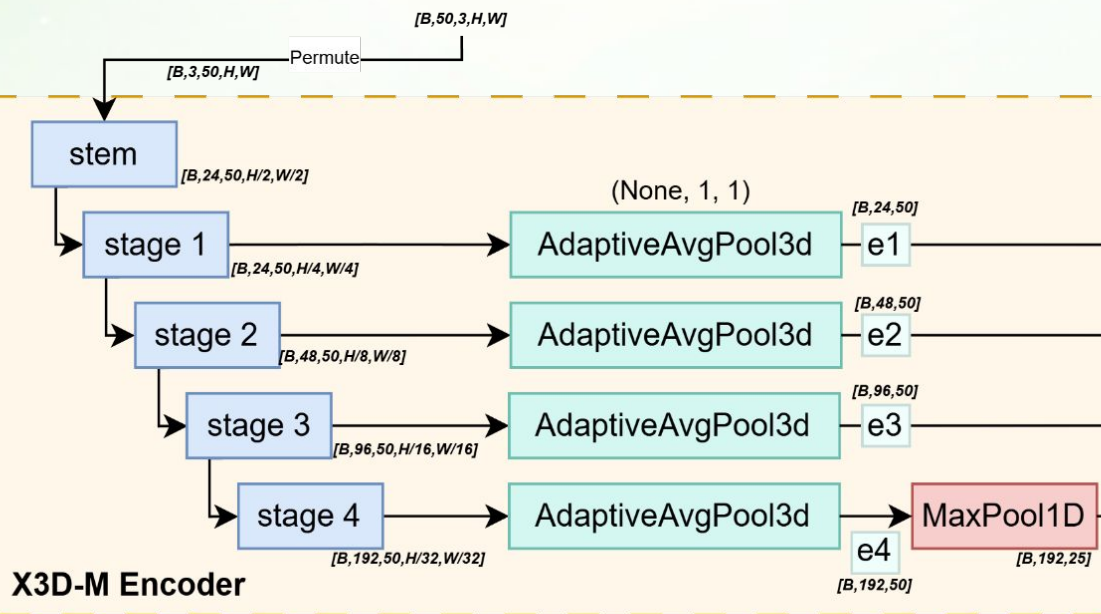
	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
X3D + UNet + Bi-GRU	47.2	45.4	41.9	40.9
+ TGLS	42.7	40.1	39.4	37.4
+ NMS	43.2	38.2	38.9	34.7

Results

- Unexpectedly, **both** techniques yield **worse results** than just the best model obtained.

Best Model

Gerard Vilaplana



Conv1D* = Conv1D → BN1d → ReLU



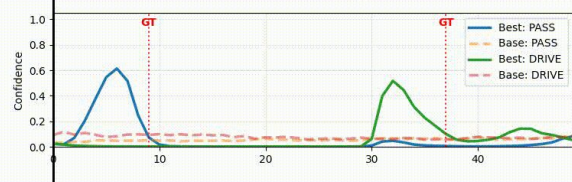
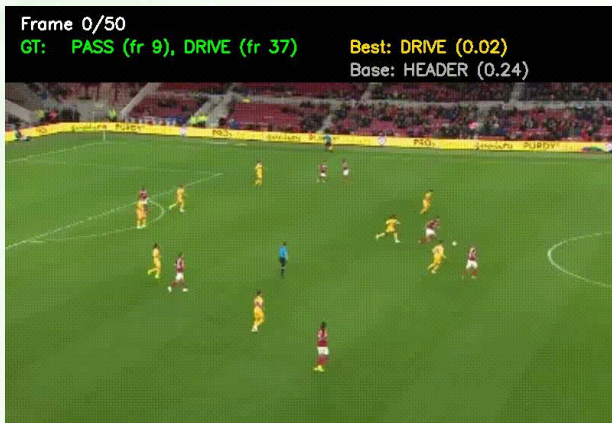
				Tolerance 1s	
		Best Model	Ablation	mAP 10	mAP 12
1&2	X3D (+ Bi-GRU)	✗	Adding Bi-GRU to X3D improves performance due to better temporal modeling and bidirectional context. X3D extracts more relevant frame-level features compared to the previous RNY008 + TCN neck setup.	-	-
3&4	X3D + UNet (+ Bi-GRU)	UNet	The initial U-Net configuration does not outperform the baseline X3D model .	43.7	41.7
5	Stride & Clip Length	2 stride + 50 CL	Stride 2 and clip length 50 provide the best trade-off between training stability and performance .	-	-
6&7	Max vs Avg Pooling	Max + GRU 2-layers 512 Hidden size	AVG limits GRU learning , most setups perform similarly . MAX shows higher variability , achieving both best and worst results.	47.2	45.4
8	TGLS & NMS	✗	No improvement, predictions are already well-localized . No improvement, low redundancy makes suppression unnecessary .	-	-

	Metrics					
	Tolerance 1s		Tolerance 0.5s		Params (M)	MACs (G)
	mAP 10	mAP 12	mAP 10	mAP 12		
Best Model W6	35.5	34.2	30.2	29.7	10.789	71.720
Best Model W7	47.2	45.4	41.9	40.9	9.099	28.421

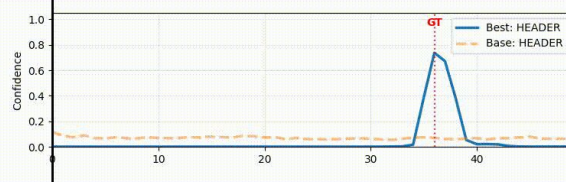
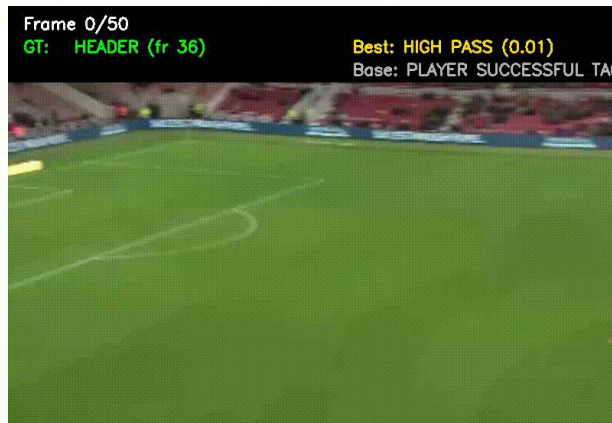
Class	Tolerance 1s	
	Baseline	Best
	mAP10	mAP10
Pass	80.55	77.19
Drive	71.29	68.46
Header	46.64	52.49
High Pass	62.53	67.90
Out	8.26	19.08
Cross	28.17	65.29
Throw In	27.20	59.91
Shot	22.32	39.66
Ball Player Block	7.23	13.42
Player Successful Tackle	1.52	9.46
Free Kick	54.54	54.54
Goal	0	18.18

Qualitative

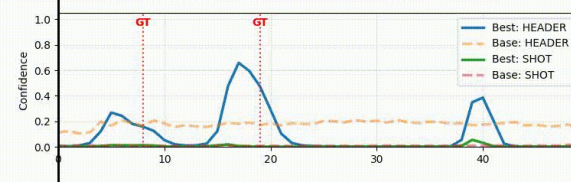
Maiol Sabater



The most frequent



Visually clear



The most difficult

Conclusions

- ✦ **X3D-M** with 3D convolutions **outperforms RNY008+TCN** by capturing spatiotemporal features natively, achieving mAP10 **47.2** vs **35.5** (+33%).
- ✦ The **UNet decoder** with temporal reduction ($L \rightarrow L' \rightarrow L$) does not surpass plain **X3D+GRU** in mAP10@1s, but shows competitive precision at 0.5s tolerance, suggesting the bottleneck improves localization at the cost of some semantic performance.
- ✦ The **bidirectional GRU** is the **most critical** component of the entire pipeline. Without it, neither X3D base nor UNet achieve competitive results. The **UNet decoder** only **adds value** when accompanied by **explicit time-domain modeling**.

Future Work

Longer temporal context

Clip Lenght =100
Stride = 2

Regularization to avoid the overfitting observed in Exp5.

Improved bottleneck design

Learnable temporal downsampling → strided Conv1D

instead of fixed MaxPool to let the model decide what to compress.

Better supervision for rare classes

Optical flow **V** Pose estimation

Visually similar actions:
CROSS vs HIGH PASS